

Exam Date & Time: 27-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov/ Dec -2024

Applications of AI in Medicine [CSE-4014]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Discuss the concept of the margin in Support Vector Machines and explain how it is maximized in the training process. (5)
 - A)
 - B) Discuss the steps to be performed in K means clustering. (5)
- 2) Demonstrate the synthetic skin lesions with its attributes and values. (5)
 - A)
 - B) Demonstrate the steps involved in minimization of error by backpropagation algorithm. (5)
- 3) Discuss the following multiclass methods: (5)
 - (i) OvsO
 - (ii) ECOC
 - A)
 - B) Explain the working process of electroencephalogram. (5)
- 4) Discuss the following liquid base cytology technologies: (5)
 - (i) SurePath
 - (ii) ThinPrep
 - A)
 - B) Discuss the deep belief networks with a neat diagram. (5)
- 5) With a neat diagram, discuss the text rank algorithm. (5)
 - A)
 - B) Design the diagnostic decision tree for the symptom "face pain". (5)

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